A PROJECT REPORT ON

CRYPTOGENESIS AND BLOCKCHAIN CREATION

SUBMITTED TO THE SAVITRIBAI PHULE UNIVERSITY, PUNE
IN THE PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF THE DEGREE

OF

BACHELOR OF ENGINEERING (Computer Engineering)

SUBMITTED BY

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2024-2025

CERTIFICATE

This is to certify that the project report entitled

CRYPTOGENESIS AND BLOCKCHAIN CREATION

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is a bonafide student of this institute and the work has been carried out by him/her under the supervision of **Prof. A.V.Sagare** and it is approved for the partial fulfillment of the requirement of Savitribai Phule Pune University, for the award of the degree of **Bachelor of Engineering** (Computer Engineering).

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ACKNOWLEDGEMENT

It gives us great pleasure in presenting the project report on ***'CRYPTO-GENESIS AND BLOCKCHAIN CREATION'****.*

We would like to express our sincere gratitude to our mentor and project guide, ***Prof. A.V.Sagare****, for providing the problem statement that forms the foundation of this project. Their guidance and support have been invaluable throughout the research and development phases of this project. We are deeply thankful for their continuous encouragement, insightful feedback, and valuable suggestions that greatly contributed to shaping the direction of this work.*

We are also thankful to our Head of the Department, ***Dr. G.V. Kale****, for their indispensable support and timely feedback throughout the project.*

Additionally, we would like to acknowledge the collaboration and support from our peers, which significantly enhanced the overall completion of the project.

This project would not have been possible without the resources and infrastructure provided by the institution, and the feedback-driven model improvement system incorporated into the project has been a key factor in ensuring its success.

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ABSTRACT

This research project, ***CRYPTOGENESIS AND BLOCKCHAIN CREATION****, begins with a comprehensive exploration of consensus algorithms that form the backbone of blockchain networks. We analyzed both classical mechanisms such as Proof of Work (PoW), Proof of Authority (PoA), and Proof of Activity with Computational Node (PoACN), as well as more advanced and experimental consensus models. Based on this study, we designed and implemented a custom blockchain tailored for use in decentralized finance (DeFi) applications and smart contract deployment, with a focus on achieving scalability, decentralization, and improved energy efficiency.*

Building upon the implemented blockchain, we further explored ways to enhance its security in the face of emerging quantum computing threats. We researched and experimented with post-quantum cryptographic techniques, including SPHINCS+, BLAKE3, Haraka, and SHA-based algorithms, to evaluate their integration and effectiveness within our system. In parallel, efforts were made to reduce the environmental footprint of the blockchain by leveraging energy-efficient cryptographic primitives and consensus strategies. The result is a secure, sustainable, and quantum-resilient blockchain framework designed to address future challenges in decentralized systems.

In conclusion, CRYPTOGENESIS AND BLOCKCHAIN CREATION represents a research driven project enriched with practical outcomes. By combining theoretical analysis with real world implementation, the project addresses key issues in blockchain development such as consensus efficiency, quantum resistance, and sustainability while delivering a working prototype that reflects the feasibility of these innovations in actual systems.

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CHAPTER 1

INTRODUCTION

1.1 Overview

Cryptogenesis and blockchain creation is a research focused project aimed at building a custom blockchain that addresses key challenges in scalability, decentralization, sustainability, and security. The project explores a range of consensus algorithms from traditional models like PoW, PoA, and PoACN to advanced experimental mechanisms and implements a blockchain tailored for DeFi and smart contract use. To future-proof the system, post quantum cryptographic techniques such as SPHINCS+, BLAKE3, Haraka, and SHA-based algorithms are integrated, alongside energy efficient strategies to reduce environmental impact, resulting in a secure and practical blockchain prototype.

1.2 Motivation

The motivation behind cryptogenesis and blockchain creation stems from the rapidly evolving landscape of blockchain technology, where scalability, security, and sustainability remain key challenges. As blockchain adoption grows, especially in sectors like decentralized finance (DeFi) and smart contracts, existing systems struggle with energy inefficiency and vulnerabilities to emerging technologies like quantum computing. This project aims to explore and address these critical issues by investigating and implementing more efficient consensus algorithms and quantum-resistant cryptographic techniques. By combining theoretical insights with practical implementation, the goal is to build a blockchain that not only performs effectively in today's decentralized systems but is also future-proof and environmentally sustainable, ensuring its long-term relevance and impact.

1.3 Problem Definition and Objectives

The project addresses two interrelated challenges in modern blockchain and decentralized systems:

1.3.1 Research on consensus mechanism

The current consensus mechanisms in blockchain, such as Proof of Work (PoW), are often criticized for their high energy consumption and environmental impact. As blockchain adoption increases, the need for more sustainable and energy efficient alternatives becomes critical. This project aims

to investigate and develop consensus algorithms that reduce the ecological footprint of blockchain systems, without compromising on scalability, decentralization, or security.

1.3.2 Creation of blockchain and research on quantum proof cryptography

As quantum computing advances, the cryptographic foundations of current blockchain systems face potential threats. Traditional cryptographic methods may become vulnerable to quantum algorithms capable of breaking them. This part of the project focuses on creating a robust blockchain while exploring post quantum cryptographic solutions, including SPHINCS+, BLAKE3, Haraka, and SHA-based algorithms, to ensure that the blockchain remains secure and resilient against future quantum threats.

1.4 Project Scope & Limitations

1.4.1 Project Scope

The proposed research project based,*Cryptogenesis and blockchain* creation, encompasses the exploration and development of a blockchain system that addresses critical challenges in scalability, energy efficiency, security, and quantum resistance. The project begins with a comprehensive study of various consensus algorithms, focusing on both classical models such as Proof of Work (PoW) and Proof of Authority (PoA) and advanced mechanisms aimed at reducing energy consumption. Based on this research, the project implements a custom blockchain optimized for decentralized finance (DeFi) applications and smart contracts. The system is designed to be decentralized and scalable while prioritizing energy efficiency to minimize its environmental impact.

A key aspect of the project's scope is the research and integration of post-quantum cryptographic techniques to ensure that the blockchain can withstand the security threats posed by future advancements in quantum computing. This includes testing algorithms such as SPHINCS+, BLAKE3, Haraka, and SHA-based methods within the blockchain infrastructure. Additionally, the project explores ways to ensure that the blockchain remains viable and sustainable in real-world applications, addressing the need for both efficient consensus mechanisms and quantum-resistant security.

The scope extends to building a fully functional blockchain prototype that incorporates these principles and validates the theoretical research

through practical implementation. The project aims to deliver a solution that can serve as a foundation for future blockchain systems that prioritize sustainability, security, and long-term adaptability.

1.4.2 Limitations in Existing Systems

While cryptogenesis and blockchain creation addresses significant challenges in blockchain technology, there are several limitations that frame the scope of the research. One of the primary limitations is the restricted focus on specific consensus algorithms. Although the project explores both traditional and advanced consensus models, it does not cover every possible consensus mechanism. Some consensus mechanisms may have been omitted due to complexity, lack of resources, or focus on more feasible alternatives in terms of scalability and environmental impact. Additionally, while energy-efficient consensus algorithms are explored, it is challenging to fully evaluate their impact in diverse real-world scenarios due to the complexity and variability of different use cases.

Another limitation is the project's focus on specific quantum-resistant cryptographic techniques, such as SPHINCS+, BLAKE3, Haraka, and SHA-based algorithms. While these methods are promising, the research does not explore every post-quantum cryptography option available, and other potentially more effective algorithms may exist. The integration and testing of these cryptographic methods within the blockchain framework are based on the current state of research and available resources, meaning future developments in quantum computing may necessitate additional revisions or enhancements.

Furthermore, the environmental sustainability aspect of the project is primarily theoretical. While energy-efficient consensus mechanisms are implemented and tested, the actual reduction in environmental footprint will depend on real-world deployment and may be influenced by factors outside the scope of this research, such as the geographic distribution of blockchain nodes and electricity sources. Finally, the blockchain created in this project is primarily a proof-of-concept prototype and is not designed for commercial use or large-scale deployment, meaning its practical utility is limited to research purposes and academic evaluation.

1.5 Methodologies of Problem Solving

The project *cryptogenesis and blockchain creation* follows a structured and research-driven approach to address the challenges outlined in the problem

statement. The entire process is divided into two major research and implementation tracks—one focused on developing energy-efficient consensus algorithms and the other on building a quantum-resilient blockchain system.

1. Research and Analysis of Consensus Mechanisms

- A detailed literature review was conducted to study existing consensus algorithms including Proof of Work, Proof of Authority, and Proof of Activity with Computational Node.
- Advanced and experimental consensus models were explored to understand their potential in improving energy efficiency and scalability.
- Comparative analysis was performed on selected algorithms based on parameters such as throughput, decentralization, fault tolerance, and environmental impact.
- Based on the findings, the most suitable mechanism was selected and used as a foundation for implementing the custom blockchain system.

2. Blockchain Design and Implementation

- A blockchain framework was developed with modules for block validation, transaction handling, and smart contract execution.
- The design was tailored to support decentralized finance applications, ensuring compatibility with programmable transaction logic.
- Emphasis was placed on reducing resource consumption while maintaining the integrity and performance of the blockchain network.

3. Post-Quantum Cryptography Integration

- Post-quantum cryptographic schemes such as SPHINCS+, BLAKE3, Haraka, and SHA variants were selected based on current research and NIST recommendations.
- Algorithms were integrated into the blockchain's signing and verification processes.
- Security testing and performance evaluation were conducted to assess resistance to quantum computing threats.

4. Validation and Testing

- The system was tested in simulated environments for functional correctness, energy consumption, and cryptographic strength.
- Outcomes were compared with traditional blockchains to assess improvements in sustainability and security.

This multi-stage methodology ensures that both theoretical insights and practical outcomes are aligned to deliver a robust blockchain solution.

CHAPTER 2

LITERATURE SURVEY

2.1 A Comprehensive Survey on Green Blockchain: Developing the Next Generation of Energy Efficient and Sustainable Blockchain Systems

This paper provides an extensive analysis of blockchain components with a focus on reducing energy consumption. It examines consensus mechanisms, network architecture, data storage, smart contract execution, mining, and block creation. The authors propose strategies to enhance energy efficiency, such as optimizing consensus algorithms and improving hardware and software components. The study serves as a guideline for researchers aiming to develop sustainable blockchain solutions.

2.2 Systematic Survey on Energy Conservation Using Blockchain for Sustainable Computing

This research offers a comprehensive analysis of blockchain architectures aimed at enhancing computing sustainability. It emphasizes scalability, resource utilization, transparency, and energy conservation. The study evaluates various decentralized structures and consensus principles to redefine computing infrastructure. The findings highlight best practices for optimizing the environmental impact of blockchain technology.

2.3 Energy Efficiency in Blockchain Networks: Analyzing Power Consumption of Consensus Mechanisms and Hash Functions

This paper analyzes critical factors contributing to power consumption in blockchain networks, focusing on consensus mechanisms and hashing techniques. It evaluates the energy efficiency of various consensus algorithms and hash functions, providing insights into optimizing blockchain systems for reduced energy usage. The study offers recommendations for developing energy-efficient blockchain networks.

2.4 A Review of Consensus Mechanisms and their Energy Consumption

This review explores the energy consumption of various consensus mechanisms used in blockchain technology. It compares traditional and modern consensus algorithms, assessing their energy efficiency and suitability for sustainable blockchain systems. The paper provides a framework for developing environmentally friendly blockchain solutions.

2.5 The Energy Footprint of Blockchain Consensus Mechanisms Beyond Proof-of-Work

This study examines the energy consumption of different blockchain consensus mechanisms, extending beyond proof-of-work. It analyzes the energy requirements of various consensus algorithms, highlighting the need for energy-efficient alternatives. The paper emphasizes the importance of selecting appropriate consensus mechanisms to reduce the environmental impact of blockchain technology.

2.6 A New Horizon for Energy Efficient Consensus Mechanisms

This paper explores sustainable and energy efficient alternatives to traditional proof of work consensus mechanisms. It focuses on consensus algorithms such as proof of stake and delegated proof of stake, evaluating their efficiency, security, and environmental impact. The study highlights the trade-offs between energy efficiency and network integrity, providing insights into the design of environmentally responsible blockchain systems.

2.7 A Systematic Review of Blockchain for Energy Applications

This study provides a comprehensive analysis of blockchain applications in the energy sector. It reviews 156 studies published since 2021, examining the selection of blockchain platforms and consensus mechanisms. The findings indicate that a significant number of studies proposing new consensus

methods or platforms face challenges in practical implementation. The paper emphasizes the importance of choosing appropriate blockchain solutions for energy applications.

2.8 Blockchain based Energy Consumption Approaches in IoT

This paper investigates the integration of blockchain technology with internet of things systems to enhance energy efficiency. It analyzes the fundamentals of blockchain, including its decentralized nature and consensus algorithms, and explores how these can be applied to various IoT fields. The study identifies hybrid blockchain models as suitable for energy efficiency in IoT, providing both qualitative and quantitative analyses along with future research directions.

2.9 A Survey on Quantum Safe Blockchain Security Infrastructure

This survey examines the security challenges posed by quantum computing to blockchain systems. It discusses the vulnerabilities of current cryptographic algorithms and explores quantum safe alternatives. The paper provides an overview of post quantum cryptographic solutions and their potential integration into blockchain infrastructures to ensure long term security against quantum threats.

2.10 Post Quantum Distributed Ledger Technology A Systematic Survey

This paper investigates the current state of post quantum cryptographic systems within blockchain frameworks. It begins with an overview of blockchain and quantum computing, examining their mutual influence. The study conducts a comprehensive literature review of quantum safe cryptosystems, assessing their readiness and potential for integration into distributed ledger technologies to mitigate quantum computing threats.

2.11 A Survey and Comparison of Post Quantum and Quantum Blockchains

This paper provides a comprehensive overview of the emerging field of quantum safe blockchain technologies. It differentiates between post quantum blockchains, which utilize classical cryptographic algorithms resilient to quantum attacks, and quantum blockchains, which leverage quantum computing and networks to rebuild blockchain foundations. The study examines differences in blockchain structure, security, privacy, and other key factors, concluding with a discussion on current research trends and challenges in these domains.

2.12 Towards Post Quantum Blockchain A Review on Blockchain Cryptography Resistant to Quantum Computing Attacks

This article studies the current state of post quantum cryptosystems and their application to blockchains and distributed ledger technologies. It examines the most relevant post quantum blockchain systems and their main challenges. The paper provides extensive comparisons on the characteristics and performance of promising post quantum public key encryption and digital signature schemes for blockchains, offering guidelines for future blockchain researchers and developers.

2.13 A Quantum Resistant Blockchain System A Comparative Analysis

This study presents a comparative analysis of a quantum resistant blockchain system. It evaluates the system's performance and security features, focusing on its resilience against quantum computing threats. The paper discusses the implementation of quantum resistant cryptographic algorithms and their impact on blockchain operations, providing insights into the development of secure blockchain systems in the quantum era.

2.14 Performance Evaluation of a Quantum Resistant Blockchain A Comparative Study with Secp256k1 and Schnorr

This research evaluates the performance of a quantum resistant blockchain by comparing it with traditional cryptographic algorithms such as secp256k1 and Schnorr. The study assesses the efficiency and security of the quantum resistant system, analyzing its suitability for practical blockchain applications. The findings contribute to understanding the trade-offs involved in adopting quantum resistant cryptographic solutions.

2.15 Security Analysis of Classical and Post Quantum Blockchains

This paper analyzes the security aspects of classical and post quantum blockchains. It examines the vulnerabilities of traditional cryptographic algorithms to quantum computing attacks and explores post quantum alternatives. The study provides a comparative assessment of different blockchain systems, highlighting the importance of transitioning to quantum resistant cryptographic methods to ensure long term security.

2.16 Quantum Resistance in Blockchain Networks

This study investigates the concept of quantum resistance in blockchain networks. It explores the potential threats posed by quantum computing to existing blockchain systems and evaluates strategies to enhance their resilience. The paper discusses the implementation of quantum resistant cryptographic techniques and their effectiveness in securing blockchain networks against future quantum attacks.

2.17 A Survey on Quantum Safe Blockchain Systems

This survey examines the security challenges posed by quantum computing to blockchain systems. It discusses the vulnerabilities of current cryptographic

algorithms and explores quantum safe alternatives. The paper provides an overview of post quantum cryptographic solutions and their potential integration into blockchain infrastructures to ensure long term security against quantum threats.

2.18 Blockchain Security Risk Assessment in Quantum Era Migration Strategies and Proactive Defense

This paper conducts a thorough risk assessment of transitioning to quantum resistant blockchains. It analyzes potential threats targeting vital blockchain components and offers a hybrid migration strategy to facilitate a smooth transition from classical to quantum resistant cryptography. The study provides actionable guidance for designing secure and resilient blockchain ecosystems in the quantum computing era.

2.19 Enhancing Blockchain Consensus Mechanisms

This research explores advancements in blockchain consensus mechanisms. It analyzes innovations aimed at improving the efficiency and security of consensus algorithms. The study discusses the implications of these advancements and their potential impact on the future of blockchain technology, providing insights into the development of robust and scalable consensus mechanisms.

2.20 NewHope Cryptographic Protocol Designed to Resist Quantum Computer Attacks

This article discusses NewHope, a cryptographic protocol designed to resist quantum computer attacks. It is based on the ring learning with errors problem and has been selected as a candidate in the NIST post quantum cryptography standardization process. The paper highlights NewHope's role in enhancing the security of blockchain systems against quantum threats.

CHAPTER 3

SOFTWARE REQUIREMENTS SPECIFICATION

3.1 Assumptions and Dependencies

3.1.1 Assumptions

- Technical Expertise: It is assumed that the project team possesses the necessary technical expertise in blockchain technology, cryptography, and software development to create a custom blockchain wallet.
- Blockchain Platform: The project assumes the selection of a specific blockchain platform (e.g., Ethereum, Binance Smart Chain, or a custom blockchain) as the underlying technology for the wallet.
- Stakeholder Engagement: The report assumes that stakeholders are actively engaged in the project, providing feedback and approvals as needed throughout the development process.
- Security Requirements: It is assumed that security is a paramount concern, and best practices for securing the wallet, including encryption and secure key management, are being followed.
- Compliance and Legal: The project assumes that the wallet and its operations comply with relevant legal and regulatory requirements, such as antimoney laundering (AML) and know your customer (KYC) regulations.
- Resource Availability: The project assumes the availability of necessary resources, including hardware, software, and skilled personnel.

3.1.2 Dependencies

- Blockchain Network Development: The creation of a custom blockchain wallet is dependent on the prior development and availability of the underlying blockchain network. The network must be operational and stable for wallet integration.
- Smart Contracts: If the wallet relies on smart contracts for certain functions, their development and deployment are a critical dependency.
- Third-Party Services: Dependencies may exist on third-party services for features like cryptocurrency exchange rates, transaction validation, and blockchain data retrieval.
- External APIs: If the wallet integrates with external systems or services, the availability and stability of those APIs are dependencies.

- Regulatory Changes: The project may be dependent on the stability of regulatory requirements, and any changes in the legal landscape could impact the wallet's operations.
- Hardware/Infrastructure: The project may depend on the availability and reliability of hardware and infrastructure, especially if the wallet is designed for specific devices or platforms.
- Testing Environments: Dependencies exist on the availability of test environments and sandboxes for thorough testing of the wallet's functionality and security.
- User Feedback: The project depends on receiving user feedback for improvements and feature enhancements. The timing and quality of feedback can impact the development roadmap.
- Project Schedule: Dependencies exist on the project schedule and timelines, including milestone achievements and delivery schedules.

3.2 Functional Requirements

3.2.1 Network and Connectivity Requirements

The project necessitates a robust and reliable network infrastructure to support the custom cryptocurrency ecosystem. This includes setting up secure and high-speed internet connectivity, optimizing server configurations, and ensuring failover mechanisms for uninterrupted service. Scalability and load-balancing solutions are essential to handle potential increases in network traffic and user transactions. Additionally, the network architecture must be designed to support peer-to-peer communication among nodes, data synchronization, and efficient transaction processing. A well-established network and connectivity framework are fundamental to the success and efficiency of the cryptocurrency project.

3.2.2 Application Requirements

- 1. User Authentication.
- 2. User Wallets: Create user-friendly wallets for various platforms (desktop, web, mobile) to enable users to store, send, and receive the custom cryptocurrency

3.3 External Interface Requirements

3.3.1 User Interfaces

- **Web-based Interface Using React:** Implement a user-friendly web interface using React.js, providing intuitive navigation and interactive features for users to access and manage their blockchain accounts, interact with smart contracts, and perform cryptocurrency transactions.
- **MetaMask Integration:** Integrate MetaMask browser extension as a user interface component, enabling users to securely manage their Ethereum accounts, sign transactions, and interact with decentralized applications (DApps) seamlessly within the web interface.
- **Alchemy API Integration:** Incorporate Alchemy API for enhanced blockchain data access and real-time updates, enriching the user interface with dynamic content and improving overall user experience.

3.3.2 Hardware Interfaces

- No Specific Hardware Requirements: Since the project primarily involves software development for blockchain and cryptocurrency applications, there are no specific hardware dependencies or interface requirements. The system should be compatible with standard computing hardware and operating systems commonly used by developers and end-users.

3.3.3 Software Interfaces

- Node.js Backend: Utilize Node.js as the backend server technology to handle HTTP requests, interact with the blockchain network, and execute business logic for processing transactions, managing accounts, and interacting with smart contracts.
- EthereumGo (Go-Ethereum): Integrate EthereumGo (geth) client software as the Ethereum node implementation to connect to the Ethereum blockchain network, synchronize with the blockchain, and interact with Ethereum smart contracts using the JSON-RPC API.
- Solidity Smart Contracts: Develop smart contracts using Solidity programming language for defining the rules and logic governing the behavior of decentralized applications (DApps) deployed on the Ethereum

blockchain. Interact with smart contracts through the Ethereum Virtual Machine (EVM) using EthereumGo client.

- Hyperledger Besu Integration: Optionally, integrate Hyperledger Besu client software for Ethereum mainnet compatibility, allowing interoperability with Ethereum-based networks and enhancing flexibility in blockchain deployment options.

3.4 Communication Interfaces

- HTTP/REST APIs: Implement HTTP/REST APIs for communication between frontend and backend components, enabling data exchange and interaction with blockchain networks, smart contracts, and external services.
- WebSocket for Real-time Updates: Utilize WebSocket communication protocol to facilitate real-time updates and notifications to users, such as transaction confirmations, account balances, and blockchain events, enhancing the responsiveness and interactivity of the user interface.
- MetaMask Integration via JSON-RPC: Communicate with MetaMask browser extension using the JSON-RPC protocol over HTTP or WebSocket connections, enabling secure interaction with Ethereum accounts and transactions directly from the web interface.
- Alchemy API for Blockchain Data: Utilize Alchemy API for accessing blockchain data, such as transaction history, smart contract events, and account information, through HTTP endpoints, providing rich and upto-date blockchain data for analysis and visualization within the user interface.

3.5 Non-Functional Requirements

3.5.1 Performance Requirements

- Transaction Throughput: Specify the number of transactions per second (TPS) your blockchain should be able to handle. This is essential for determining the capacity of your network.
- Latency: Define the acceptable transaction confirmation time, which represents how quickly a transaction can be verified and added to the blockchain.

- Scalability: Outline how the blockchain will scale as more users and nodes join the network. Consider both on-chain and off-chain scalability solutions.
- Consensus Algorithm Efficiency: Evaluate the efficiency of your chosen consensus algorithm. Ensure it can maintain security while processing transactions swiftly.

3.5.2 Safety Requirements

- Data Encryption: Implement strong encryption mechanisms to protect sensitive data, such as private keys, user information, and transaction details, both in transit and at rest.
- Access Control: Define strict access control policies to restrict who can access and modify the blockchain and cryptocurrency systems. Implement authentication and authorization mechanisms.
- Smart Contract Auditing: Before deploying smart contracts, conduct thorough code audits to identify vulnerabilities and ensure they cannot be exploited by malicious actors.
- Penetration Testing: Conduct regular penetration testing to identify and address security weaknesses in the blockchain and cryptocurrency systems.

3.5.3 Security Requirements

- DDoS Mitigation: Protect the network from Distributed Denial of Service (DDoS) attacks by employing anti-DDoS solutions.
- Consensus Security: Protect the consensus mechanism from Sybil attacks, 51% attacks, and other threats.
- Redundancy and Disaster Recovery: Establish redundancy and disaster recovery plans to ensure system availability and data integrity in case of hardware failures or other disasters.
- Monitoring and Alerts: Implement real-time monitoring and alerting systems to detect and respond to unusual activities or anomalies.
- Malware Protection: Use anti-malware and antivirus solutions to protect the systems from malicious software.

3.5.4 Software Quality Attributes

- Reliability: Reliability refers to the ability of the software to perform its intended functions consistently and predictably under various conditions. Reliable software should minimize the occurrence of failures, errors, and unexpected behavior, ensuring high availability and fault tolerance.
- Performance: Performance relates to the efficiency and responsiveness of the software in executing tasks within acceptable time frames and resource constraints. It includes metrics such as response time, throughput, scalability, and resource utilization, aiming to optimize system performance to meet user expectations and workload demands.
- Security: Security encompasses measures to protect the software system and its data from unauthorized access, manipulation, and malicious attacks. It includes features such as authentication, authorization, encryption, data integrity, and vulnerability management, ensuring confidentiality, integrity, and availability of sensitive information.
- Testability: Testability refers to the ease with which the software can be tested to validate its functionality, performance, and reliability. It includes factors such as test coverage, observability, controllability, and repeatability, facilitating effective testing and quality assurance processes to identify and address defects and deficiencies.

3.6 System Requirements

3.6.1 Software Requirements

- Frontend: React.js JavaScript
- Backend: Node.js, Express.js
- Blockchain: Modern Javascript, SHA-256, Solidity
- Web3 API, messaging- pub/sub , Redis or pubNub

3.6.2 Hardware Requirements

- 8 GB RAM
- 1 TB Hard Disk

- Intel Core i3 Processor
- 4 Core CPU

3.7 Analysis Models: SDLC Model to be applied

The Agile model is selected for this project due to its flexibility, adaptability, and emphasis on continuous improvement. This development approach is particularly well-suited for building a dynamic Anomaly Detection System that evolves with user interaction and environmental changes.

3.7.1 Rationale for Agile Selection

- **Iterative Development:** The system is developed in short, manageable sprints, allowing for incremental delivery of key features such as file categorization, real-time logging, model-based anomaly detection, and explainability using LLMs.
- **Frequent Feedback Loops:** Regular interaction with end-users and stakeholders ensures that insights and feedback are incorporated into each sprint cycle, resulting in a more user-aligned solution.
- **Modular Integration:** Machine learning models (e.g., Autoencoder, Isolation Forest, LOF, One-Class SVM) are developed and validated in independent modules, making them easier to test, refine, and deploy.
- **Continuous Improvement:** Each iteration includes:
  - New data collection from live file operations
  - Feature re-engineering and model retraining
  - Feedback-driven enhancements to system UI and functionality
  - Integration of updated models into the backend

3.7.2 Benefits of Agile for This Project

- Enables rapid adaptation to evolving user needs and data patterns
- Supports ongoing optimization of anomaly detection performance
- Aligns well with the feedback loop required for ML model explainability

- Facilitates real-world testing and validation of detection accuracy

By following Agile, the final system will be both robust and responsive, capable of adapting to changing file behaviors and threat landscapes in real-time.

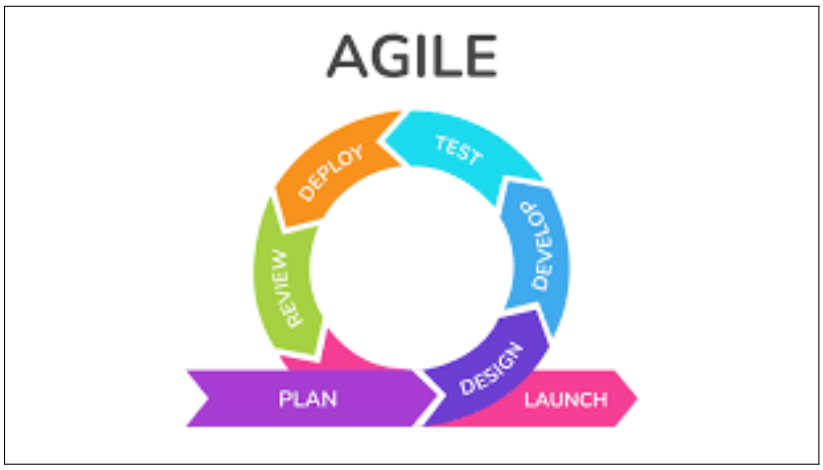


Figure 3.1: Agile model

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

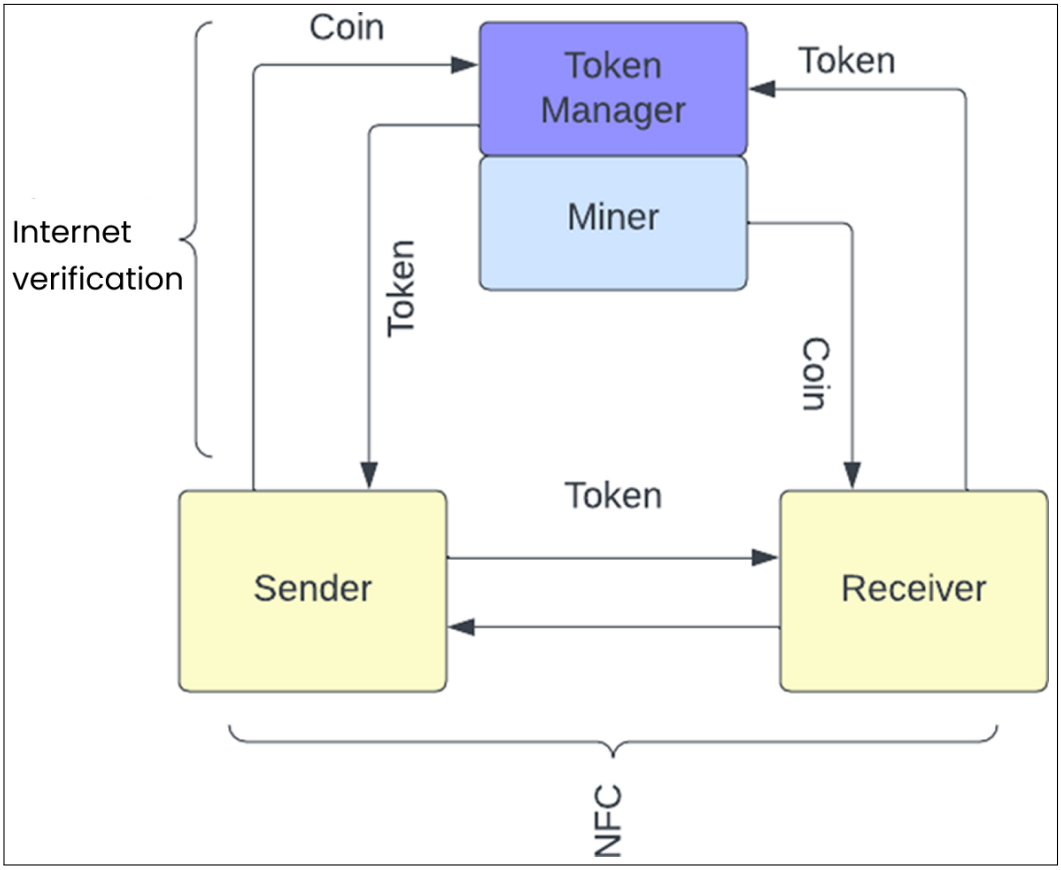


Figure 4.1: Pure Wallet Blockchain architecture

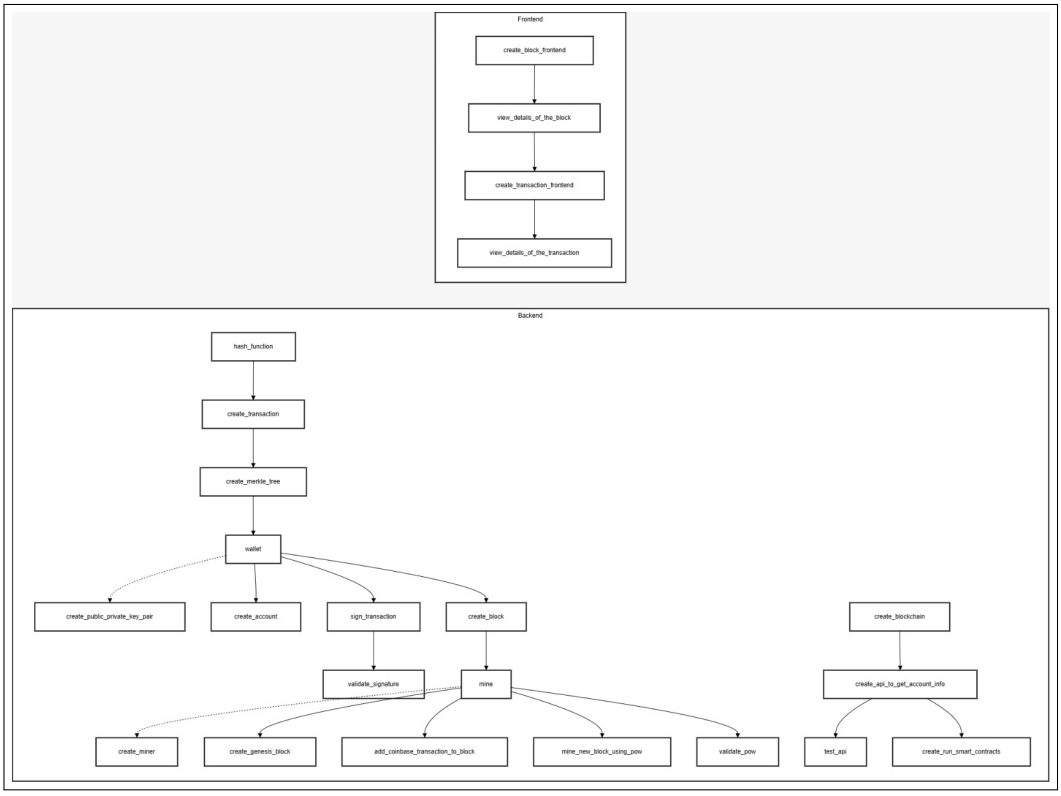


Figure 4.2: Overall high level design

The system architecture of a blockchain wallet is designed to provide secure and decentralized storage for digital assets, primarily cryptocurrencies. It consists of three key components: the user interface, the blockchain network, and the wallet software. The user interface allows users to interact with the wallet, manage their digital assets, and initiate transactions. The blockchain network is the distributed ledger that records all transactions and ensures their immutability. The wallet software acts as a bridge between the user interface and the blockchain network, enabling users to create and manage cryptographic keys, sign transactions, and securely store their private keys. Together, these elements ensure that users have complete control over their digital assets, safeguarding their security and privacy in a trustless environment.

4.2 Data Flow Diagram

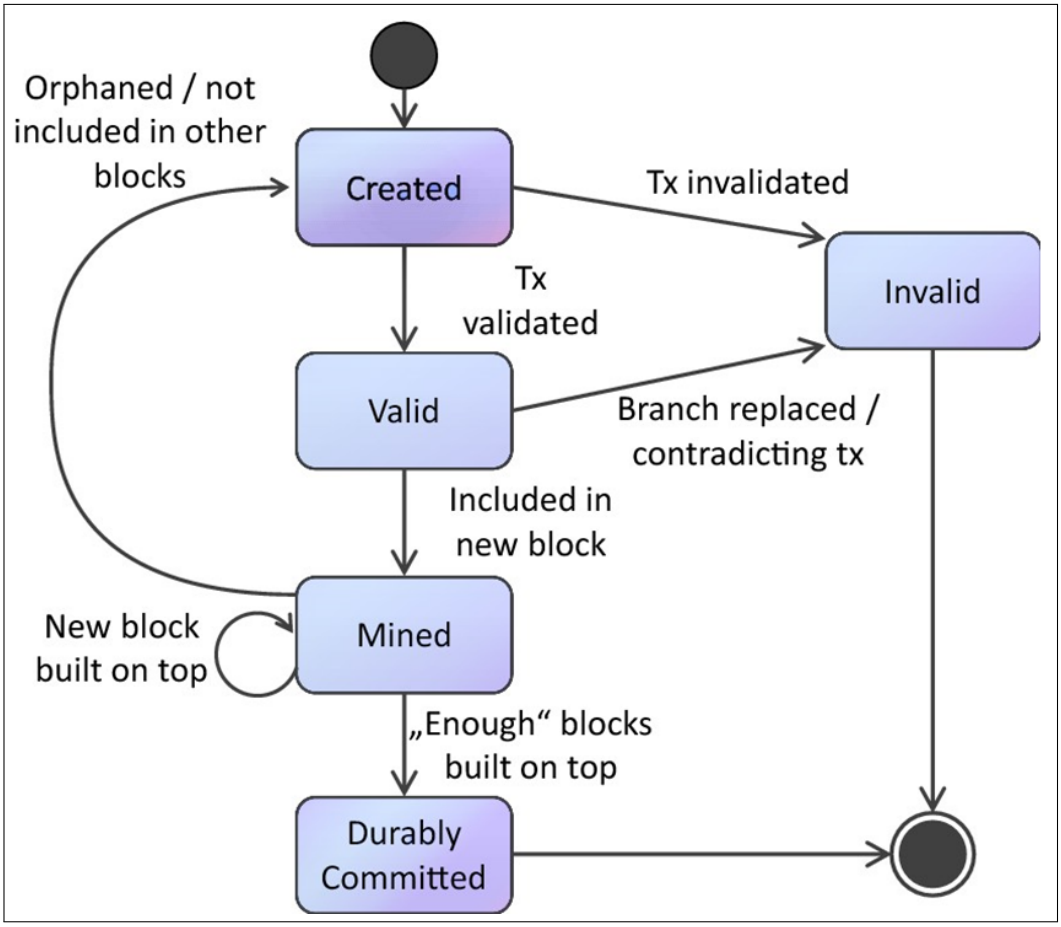


Figure 4.3: Data flow diagram

The 4.3 figure shows Data flow diagram for our application.

- 1. Transaction is initiated
- 2. Validation is done of transaction
- 3. New block is built on top by mining
- 4. Committed to ledger and send to every node in Blockchain.

4.3 UML Diagrams

This UML diagram outlines the architecture and key components involved in the creation of our custom blockchain and cryptocurrency. It visually represents the interactions and processes foundational to our innovative digital currency system. Creating custom blockchains and cryptocurrencies is a complex task involving a variety of technical components. UML (Unified Modeling Language) diagrams can be very helpful in planning and documenting the design and architecture of your blockchain system. Here are some key UML diagrams that you might consider creating for your project:

- 1. **Use Case Diagram:**This diagram will help you identify the different actors interacting with your system (e.g., users, miners, external services) and the various functionalities they can perform, like creating transactions, mining blocks, validating transactions, and exchanging cryptocurrency.
- 2. **Class Diagram:** This diagram will be central to the design of your blockchain. It should detail the classes you will need to implement the blockchain and the relationships between them. Classes might include Block, Transaction, Wallet, Blockchain, Node, etc.
- 3. **Sequence Diagram:**This diagram will describe how objects interact in sequence to complete transactions or mine new blocks. It helps in understanding the flow of data and control among various components of the system during the execution of specific operations.
- 4. **Activity Diagram:** This diagram shows the workflow of activities involved in various processes within the blockchain, such as the process of transaction validation or block mining.
- 5. **Component Diagram:** This diagram helps visualize the organization and dependencies among the software components that make up your blockchain system, such as databases, user interfaces, and external interfaces.
- 6. **Deployment diagram:** This will show the physical deployment of your blockchain system across various hardware nodes and how they are connected. This is important for understanding and planning the scalability and resilience of the network.

4.4 Class Diagram

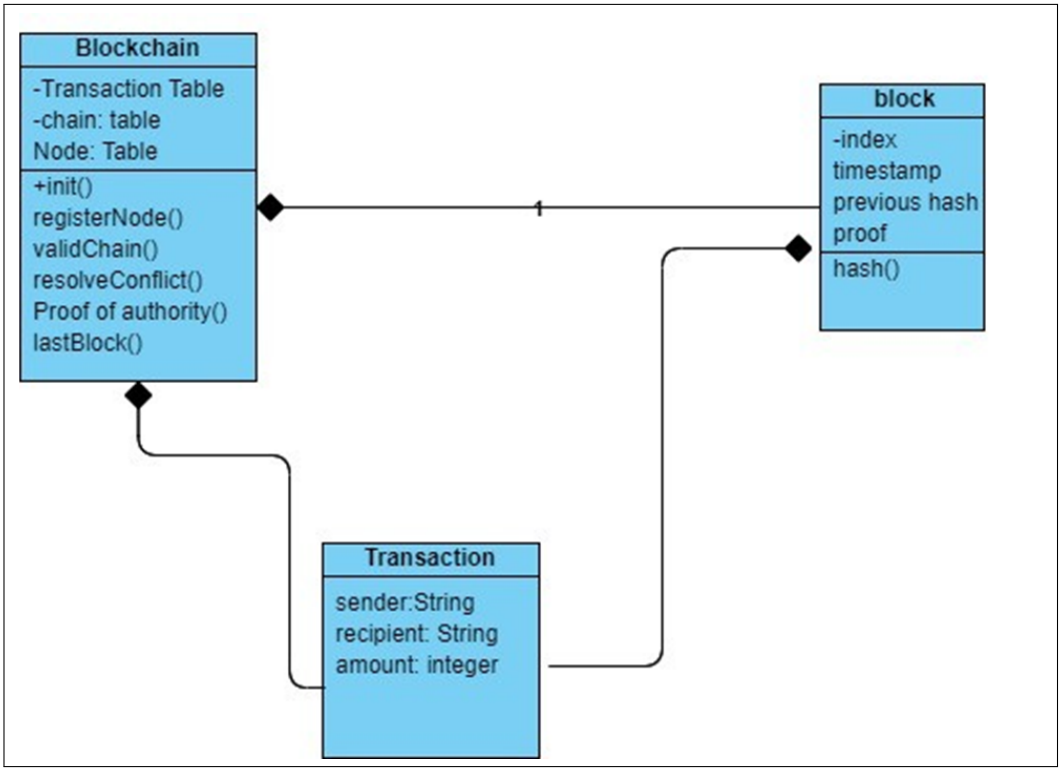


Figure 4.4: Class diagram

Below is a class diagram that illustrates the structure of our custom blockchain and cryptocurrency, detailing the essential classes and their interactions. This diagram serves as the blueprint for our system's foundational architecture.

4.5 Sequence Diagram

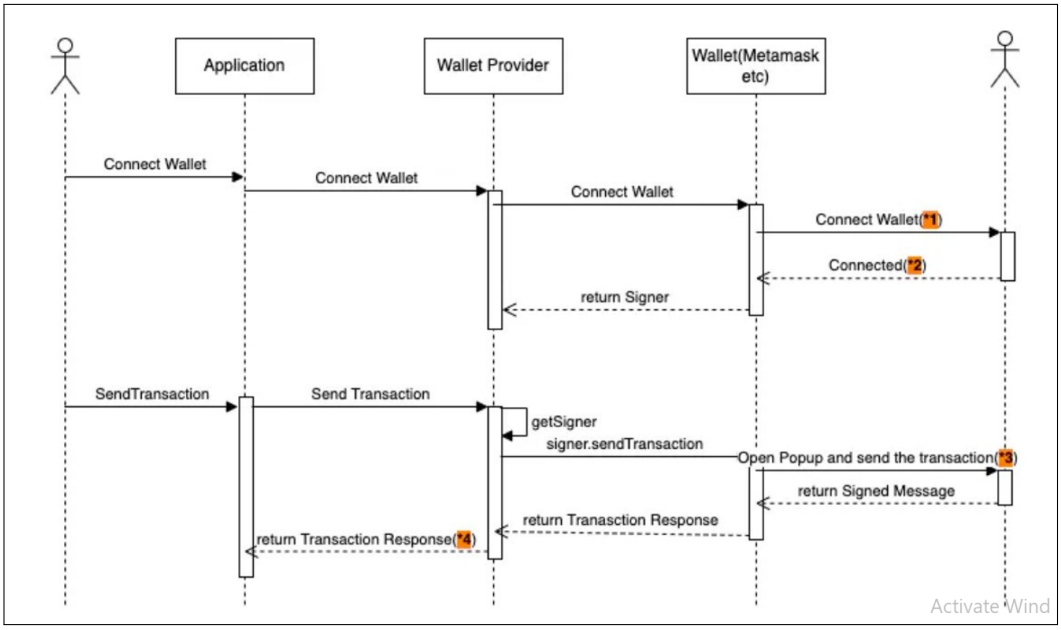


Figure 4.5: Sequence diagram

The above figure depicts the sequence diagram for our application:

- 1. Firstly the user Connects the wallet to Wallet provider.
- 2. Wallet provider connect to Blockchain network.
- 3. After that user can send transaction.
- 4. Transaction is signed and validated
- 5. response is sent for Transaction

4.6 Activity Diagram

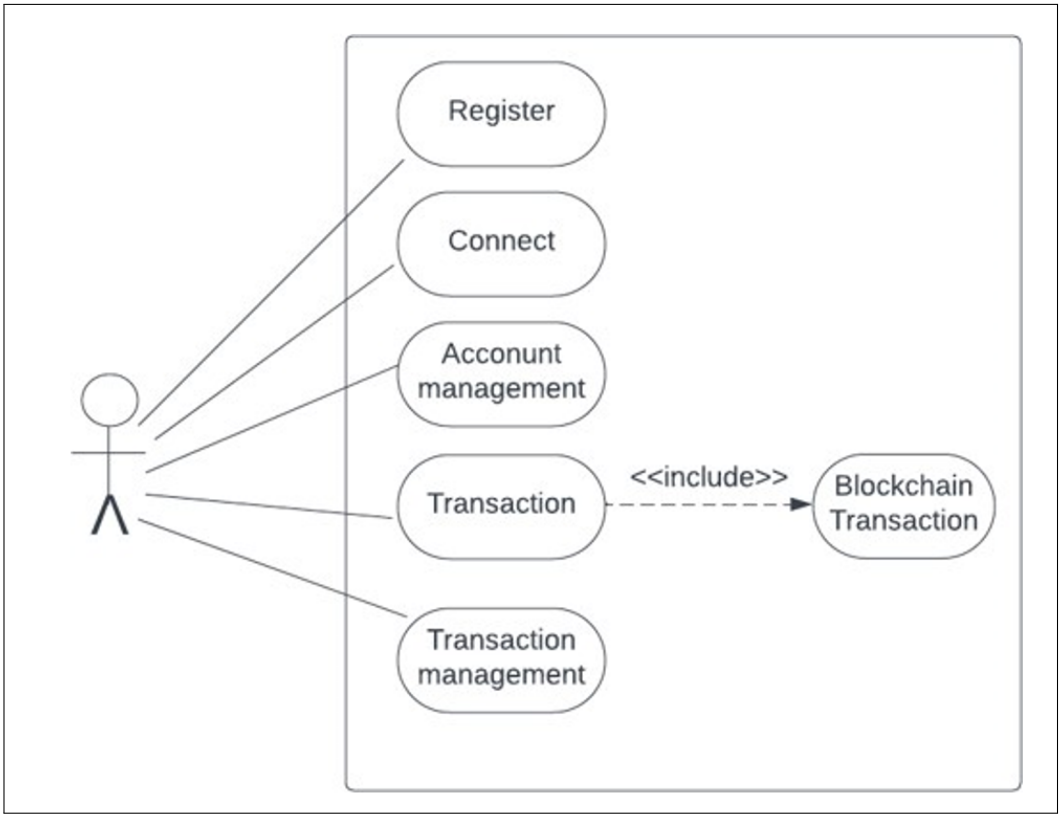


Figure 4.6: Activity diagram

4.7 Component Diagram

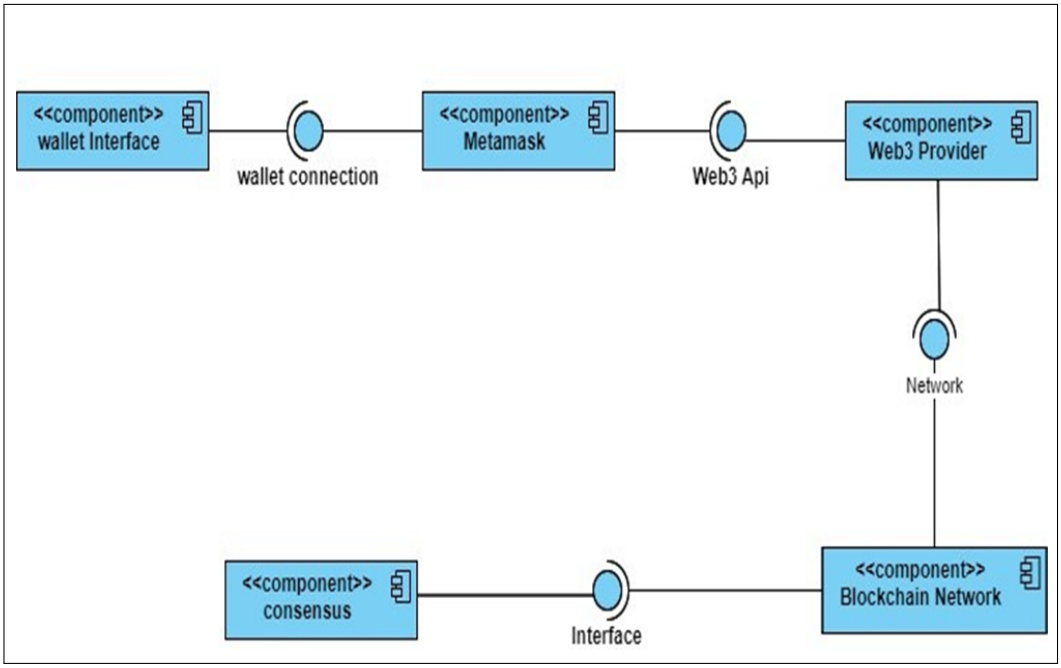


Figure 4.7: Component diagram

Our project focuses on the innovative design of a custom blockchain and cryptocurrency system. The following component diagram illustrates the architecture and interrelationships of the key components involved in this system.

4.8 Deployment diagram

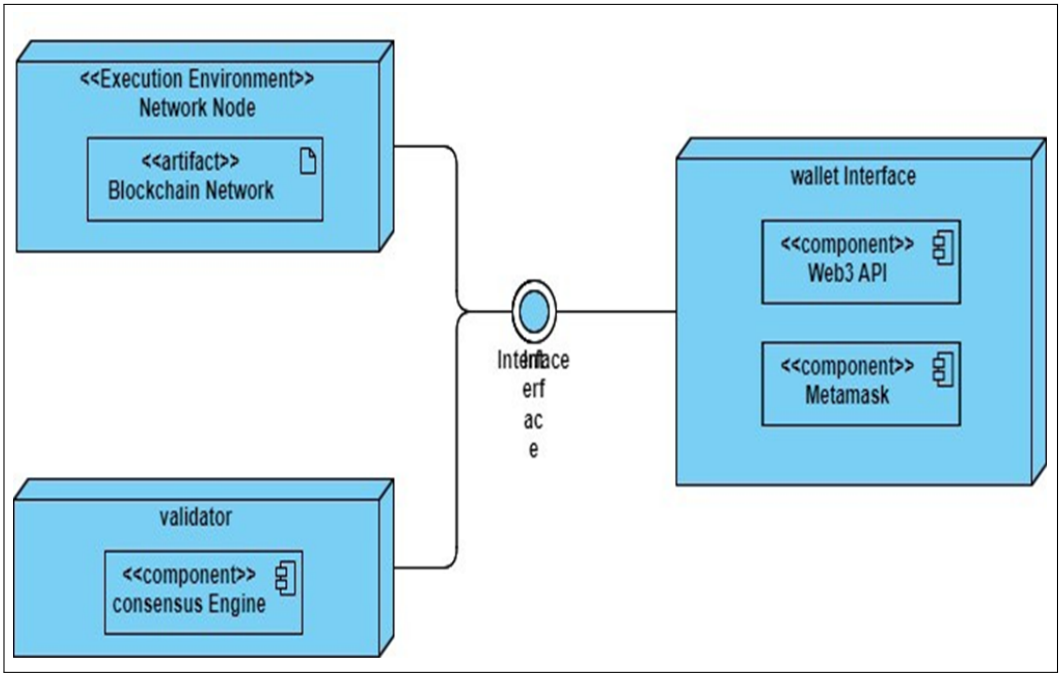


Figure 4.8: Deployment diagram

In the following deployment diagram, we illustrate the architecture of our newly developed custom blockchain and cryptocurrency system. This visualization captures the arrangement of nodes, network connections, and components essential for operation and security.

CHAPTER 5
PROJECT PLAN

5.1 Project Estimate

In the estimation phase, we have assessed the resources, time, and costs required for the creation of a custom blockchain wallet and blockchain network. This includes hardware, software, development hours, and any potential third-party services. The estimates are as follows:

- Development Time: 6 months.
- Hardware and Infrastructure: As Blockchain it requires strong and efficient hardware as in software requirement.
- Third-Party Services: Depends on Transaction and Gas fees.

5.1.1 Reconciled Estimates

After a detailed analysis of the estimates and potential risks, the reconciled estimates are as follows:

- Development Time: 4-5 months
- Transaction Costs: $0.01 / transaction-based
- Hardware and Infrastructure: $5,000 (blockchain network)
- Third-Party Services: $100

5.1.2 Project Resources

The technical resources required for the creation of a blockchain cryptocurrency are a fundamental cornerstone of the project's success. At the heart of this endeavor is the development team, comprised of blockchain experts, software engineers, and cryptography specialists. These individuals possess the necessary expertise to design and construct the blockchain network, implement consensus mechanisms, and develop smart contracts. Alongside the development team, various technical tools and software come into play. These include blockchain development frameworks and integrated development environments (IDEs) to streamline coding and testing processes. Robust technical infrastructure, including servers or cloud resources and networking equipment, is indispensable for hosting blockchain nodes and ensuring network stability. Furthermore, the project demands the vigilance of security specialists, who are tasked with safeguarding the network from cyber threats and vulnerabilities, employing encryption, network security measures,

and penetration testing tools. Development and testing environments, such as testnets and private blockchain networks, provide essential platforms for experimenting with and refining the cryptocurrency's features. A comprehensive suite of technical documentation guides both users and developers, while quality assurance and automated testing tools bolster code quality and security. Smart contract development tools like Solidity compilers and debugging environments are essential for crafting secure and efficient contracts. Data backup and recovery solutions, along with security and compliance monitoring tools, contribute to safeguarding the cryptocurrency's integrity. API integration tools allow external services to interact with the blockchain, while data analytics software aids in monitoring user adoption and network performance. If the project involves community decision-making, decentralized governance tools may also be necessary. Proper allocation and utilization of these technical resources are paramount to the successful development and operation of a blockchain cryptocurrency.

5.2 Risk Management

Managing risks is crucial for the successful implementation of our project.

5.2.1 Risk Identification

In the risk identification phase, we have identified potential risks associated with the creation of the blockchain wallet. These risks include:

- Technology Risks: Uncertainties in blockchain technology, including compatibility issues with different blockchain platforms or potential vulnerabilities.
- Security Risks: Risks related to the security of the wallet, including potential security breaches, data leaks, and unauthorized access.
- Regulatory Risks: Risks associated with changing cryptocurrency regulations and compliance requirements.
- Market Risks: Risks related to market dynamics, including shifts in user preferences and competition.
- Operational Risks: Risks related to operational challenges, such as unexpected downtime or network disruptions.

5.2.2 Risk Analysis

The risk analysis phase involves assessing the impact and likelihood of each identified risk. The analysis includes the following:

- Impact Assessment: Evaluating the potential consequences of each risk on the project, including its effect on the timeline, budget, and overall success.
- Likelihood Assessment: Determining the likelihood of each risk occurring based on historical data and expert judgment.
- Risk Prioritization: Prioritizing risks based on their potential impact and likelihood to focus on the most critical ones.

5.2.3 Overview of Risk Mitigation, Monitoring, Management

In the risk mitigation, monitoring, and management phase, we have developed strategies to address the identified risks. This includes:

- Risk Mitigation Strategies: Defining specific actions and measures to minimize the impact and likelihood of each risk.
- Monitoring and Early Warning: Implementing continuous monitoring mechanisms to detect early signs of potential risks. This includes real-time security monitoring and performance tracking.
- Risk Management Team: Establishing a dedicated risk management team responsible for assessing and addressing risks. The team will meet regularly to evaluate the project's risk status and take proactive measures.
- Contingency Plans: Developing contingency plans for high-impact risks, outlining the steps to be taken in case a risk materializes.
- Communication Plan: Defining a clear communication plan for reporting risks to stakeholders, team members, and project sponsors.
- Regular Review: Regularly reviewing and updating the risk management plan to adapt to changing project dynamics.

Effective risk management is an essential part of our project to ensure the successful development and deployment of the blockchain wallet.

5.3 Project Schedule

5.3.1 Project Task Set

- **Task 1: Project Initiation (1 week)**
  - Define project goals and objectives.
  - Assemble the project team and allocate roles and responsibilities.
  - Conduct initial research and feasibility analysis.
- **Task 2: Blockchain Design (2 weeks)**
  - Design the architectural framework for the blockchain.
  - Determine the consensus mechanism (e.g., PoW, PoS).
  - Outline data structures and network protocols.
- **Task 3: Consensus Algorithm Implementation (2 weeks)**
  - Select and implement the chosen consensus algorithm.
  - Fine-tune the algorithm for network security and performance.
- **Task 4: User Interface Integration and Node Development (8 weeks)**
  - Develop and configure blockchain nodes for transaction validation.
  - Establish the node communication protocol.
  - Ensure node synchronization with the blockchain network.
  - Develop user-friendly wallet applications and blockchain explorers.
- **Task 5: Testing and Quality Assurance (4 weeks)**
  - Conduct thorough testing, including unit testing, integration testing, and security audits.
  - – Address and rectify vulnerabilities and issues.

5.3.2 Task Network

5.3.3 Project Task Set

5.3.4 Timeline Chart

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

The proposed system is developed using a modular and scalable architecture that follows a phase-wise implementation:

- **File Categorization (Phase 1)**
- **Anomaly Detection (Phase 2)**
- **Explainable Anomalies (Phase 3)**

Each phase builds on the previous one, integrating real-time monitoring, machine learning, and explainable AI to create a robust and intelligent file management and security system.

6.1.1 Phase 1: File Categorization

This phase focuses on categorizing files based on both metadata and content using machine learning and rule-based logic.

Module 1: File Operation Logger This module performs real-time logging of file activities and collects necessary data for categorization.

Captured Attributes Include:

- File name and extension
- MIME type (verified using the magic library)
- File operation (Insert, Modify, Rename, Delete)
- Timestamp and user/process ID
- File path and size

Module 2: Categorization Engine This module classifies files into categories such as Documents, Images, Media, Code, etc.

Key Features:

- **Supervised ML Models:** Decision Tree and Random Forest trained on 20K+ labeled file extensions and MIME types.
- **Hash Data Structure:** Used for faster and more accurate categorization than ML in known cases.
- **Directory/File-Based Input:** Classification supported at both directory and individual file levels.

- **Dynamic Learning:** Unknown extensions are cached and added to the dataset once a threshold is reached, triggering retraining.
- **Feedback Loop:** Incorrect classifications are flagged and corrected by users, enhancing model accuracy over time.

6.1.2 Phase 2: Anomaly Detection

This phase is responsible for identifying abnormal file behavior through continuous monitoring and behavioral modeling.

Module 3: Anomaly Detection Engine This module monitors operations and detects deviations using advanced ML techniques.

Core Functionalities:

- **Feature Extraction:** Extracts meaningful features like operation frequency, time gaps, file sizes, and behavioral trends.
- **Model Stack (Ensemble):**
  - Isolation Forest
  - Local Outlier Factor
  - One-Class SVM
  - Autoencoder
  - River model for streaming data
- **Severity Classification:** Anomalies are tagged with severity levels (low, medium, high) based on feature deviation.
- **Offline and Parallel Execution:** Optimized for large-scale logs with multi-threaded batch processing.

6.1.3 Phase 3: Explainable Anomalies

This phase adds interpretability to the anomalies detected, making the system transparent and actionable.

Module 4: Explainability Engine Components:

- **Clustering:** DBSCAN clusters similar anomalies to reduce redundancy and improve explanation efficiency.
- **LLM Explanation:** Uses Ollama-hosted Llama3 to generate natural language descriptions of why an anomaly was flagged.

- **Explanation Storage:** All LLM outputs are stored in a CSV format and displayed in the frontend dashboard.

Module 5: Dashboard & Reporting

- **Real-Time Alerting:** Notifies users/admins about high-severity anomalies.
- **Interactive Dashboards:** Visual representation of categories, anomalies, and system trends.
- **Reports:** Weekly/monthly summary reports with anomaly trends and model metrics.

6.2 Tools and Technologies Used

The development and deployment of the proposed system leveraged a combination of modern tools, libraries, and platforms to ensure scalability, performance, and explainability. The following technologies were instrumental in different phases of the project:

6.2.1 Programming Language

- **Python 3.11:** Python was the primary language for building all modules due to its simplicity, vast ecosystem of machine learning libraries, and strong community support.

6.2.2 Machine Learning & Deep Learning Frameworks

- **Scikit-learn:** Used for implementing classical ML algorithms like Decision Trees, Random Forests, One-Class SVM, and Local Outlier Factor (LOF).
- **TensorFlow 2.x + Keras:** Used for training deep learning models such as CNNs (for content-based classification) and LSTM (for time-series anomaly patterns).
- **River:** A library for real-time machine learning on streaming data, used in the anomaly detection ensemble.

6.2.3 Explainability & Large Language Models (LLMs)

- **Ollama - Llama3 Model:** Employed for generating natural language explanations of both classification outputs and detected anomalies. Enhances system transparency and helps users understand model behavior.

6.2.4 Monitoring and File System Interaction

- **Watchdog:** A Python library that continuously monitors file system changes (creation, deletion, renaming, modification). It was used to implement the File Logger module.

6.2.5 Data Handling and Processing

- **Pandas:** Used for data cleaning, manipulation, and transformation of log data into model-ready formats.
- **NumPy:** Essential for numerical operations, especially in feature engineering and matrix operations.

6.2.6 Visualization and Reporting

- **Matplotlib & Seaborn:** Used for generating plots, behavior trend graphs, and anomaly heatmaps.
- **ReCharts:** Employed for interactive dashboards and visual reports.

6.2.7 Version Control

- **Git & GitHub:** Used for source code management, version control, and team collaboration.

6.3 Algorithm Details

Input:

- Files present in the system

- Real-time file operations (insert, delete, update, rename)
- Log directory path (C:/Users/<User_Name>/Documents/DriveLogs/logs.fs)

Output:

- Anomalies detected in file operations
- Human-readable explanations for anomalies
- Visualized and filterable dashboard output

6.3.1 Step 1: File Categorization

1. **For each file in the system:**
   - Retrieve its file extension.
   - Verify its actual content type using magic (libmagic) to extract MIME type.
2. **Map files to categories:** (Image, Audio, Video, PDF, Source Code, etc.) using a predefined hashmap.
3. **Store categorized metadata for future validation and feature enrichment.**

6.3.2 Step 2: Real-Time File Logging

1. **Start a persistent file monitoring service using Watchdog.**
2. **For each operation:**
   - Capture:
     * Timestamp
     * Filename & Path
     * File Category (from Step 1)
     * Operation Type (Insert, Update, Delete, Rename)
     * Bytes Modified (for updates)
     * New Filename (for renames)
3. **Serialize and store logs in ZODB (.fs format) for efficient, persistent, and offline-supported storage.**

6.3.3 Step 3: Trigger Anomaly Detection Pipeline

1. **The pipeline is manually triggered by the user through the frontend dashboard** (via "Run Pipeline" button).
2. **The backend loads logs from the configured .fs path in ZODB.**

6.3.4 Step 4: Feature Engineering & Preprocessing

1. **Extract and engineer the following features from logs:**
   - Numerical Timestamp (transformed from datetime)
   - File Category (categorical)
   - Operation Type (one-hot encoded)
   - Bytes Modified (numeric)
   - Operation Frequency (number of similar operations within a time window)
2. **Normalize features where required for consistency across models.**

6.3.5 Step 5: Ensemble Anomaly Detection

1. **Feed preprocessed feature vectors into the following models:**
   - **Isolation Forest:** Detects anomalies based on data isolation
   - **One-Class SVM:** Learns decision boundary for normal ops
   - **Local Outlier Factor (LOF):** Measures local density deviation
   - **Autoencoder:** Deep neural network to reconstruct patterns and detect deviations
   - **River Model:** Incremental stream learning for online anomaly detection
2. **Ensemble logic:**
   - Combine outputs via voting, score averaging, or thresholding.
   - Flag a log entry as anomalous if it crosses the combined threshold.

6.3.6 Step 6: Explainability via Local LLM

1. **For each detected anomaly:** Pass log details, features, and context into a local LLM instance (e.g., LLaMA 3 hosted on Ollama).
2. **Use a templated prompt to generate a natural language explanation:**

   > "This file rename operation was flagged because it involved a system file being renamed outside of regular hours, which is unusual for this user."

3. **Store explanation alongside the anomaly record as a key-value pair in JSON or CSV.**

6.3.7 Step 7: Visualization & Dashboard Output

1. **Final structured anomalies (with explanations) are sent to the React frontend dashboard via Flask API.**
2. **Enable advanced filters:**
   - Time Window
   - File Category
   - Operation Type
   - Severity Level / Risk Score
3. **Allow data export in CSV or JSON format.**
4. **Generate rich visualizations:**
   - Time-Series Graphs showing anomaly trends over time
   - Heatmaps of file activity and risk intensity
   - Bar/Pie Charts breaking down anomalies by category or type

CHAPTER 7

SOFTWARE TESTING

7.1 Types of Testing

Thorough testing was conducted throughout the development lifecycle to ensure the robustness, reliability, and usability of Cryptogenesis algorithms and blockchain wallets. Various testing methodologies were applied at different stages, focusing on both individual modules and the complete pipeline.

7.1.1 Unit Testing

Each module was tested independently to validate its core functionality:

- This involves testing individual components or units of the blockchain or cryptocurrency software to ensure they function correctly in isolation. For blockchain, this could include testing smart contracts, consensus algorithms, and cryptographic functions. For cryptocurrency, it could involve testing wallet functionalities, transaction processing, and security features.

7.1.2 Integration Testing

Integration tests were carried out to ensure smooth data flow and interaction between modules:

- Integration testing focuses on testing how different components of the blockchain or cryptocurrency software work together. This includes testing interactions between nodes in a blockchain network, communication between different layers of the software stack, and interoperability with external systems such as wallets and exchanges.

7.1.3 System Testing

All key features of the system were tested against their functional requirements:

- System testing evaluates the overall behavior and functionality of the blockchain or cryptocurrency software as a whole. This includes testing the performance, scalability, and reliability of the

system under various conditions, such as high transaction volumes or network congestion. It also involves testing security features to ensure protection against attacks and vulnerabilities.

7.1.4 Performance Testing

Stress testing was conducted to evaluate the system's behavior under load:

- Performance testing assesses the speed, responsiveness, and scalability of the blockchain or cryptocurrency software. This involves simulating real-world conditions and stress-testing the system to identify bottlenecks, latency issues, and performance limitations. Performance testing is essential for ensuring smooth operation under heavy loads and during peak usage periods.

7.1.5 Security Testing

Security testing in this context involved evaluating the resilience of cryptographic mechanisms and consensus protocols against potential threats, including quantum attacks and network vulnerabilities:

- Security testing is crucial for blockchain and cryptocurrency projects due to the sensitive nature of financial transactions and the potential for cyber attacks. This includes testing for vulnerabilities such as double spending, 51% attacks, smart contract bugs, and cryptographic weaknesses. Penetration testing, code reviews, and auditing are common techniques used for security testing.

7.2 Test cases & Test Results

For our creation of custom blockchain and cryptocurrency, the test cases and results are critical to ensure its success.

- The user registration and authentication processes should be resilient, enabling smooth access while upholding security standards. Actual Result: The implementation of user registration and authentication processes have been successful, enabling users to securely create accounts and login without encountering any issues.

- Transaction security, supported by blockchain technology, guarantees that all financial interactions remain tamper-proof and encrypted. Actual Result: The platform ensures transaction security through blockchain encryption, instilling users with confidence in the safety of their financial activities.
- Performance testing guarantees that models efficiently deliver accurate results. Actual Result: Performance testing confirms that AI models consistently provide accurate results within anticipated timeframes, effectively fulfilling user requirements.

Through the aforementioned testing and evaluation, our platform offers secure, transparent, and democratized access.

CHAPTER 8

RESULTS

8.1 Outcomes

The proposed system has been comprehensively evaluated across four core components—file categorization, anomaly detection, LLM-based explanation, and real-time visualization. The evaluation highlights the robustness, accuracy, and practicality of the implemented solution in a real-world file operation monitoring environment.

8.1.1 File Categorization Engine

The intelligent file categorization component combines metadata analysis, supervised machine learning, and MIME-type-based hashing for accurate classification. It supports over 20,000 unique file extensions and more than 400 defined categories.

Table 8.1: Performance Evaluation of File Categorization Models

Model	**Accuracy** (%)	**Precision** (%)	**Recall** (%)	**F1-Score** (%)
Decision Tree	89.3	88.7	87.9	88.3
Random Forest	93.6	94.1	92.8	93.4
Hash-Based Mapping	99.0	99.0	99.0	99.0

The hash-based MIME mapping provides the highest accuracy with minimal overhead and is adopted as the primary classification method, whereas machine learning models are used for unknown or dynamically observed file extensions.

8.1.2 Anomaly Detection Engine

This module monitors real-time file operations including insertions, updates, deletions, and renaming. Logs are stored using a persistent object database (ZODB), and multiple anomaly detection algorithms are used in an ensemble configuration to identify suspicious behaviors.

The ensemble method delivers the most balanced performance, outperforming individual models in both accuracy and false positive rate while maintaining reasonable execution time.

Table 8.2: Evaluation of Anomaly Detection Models

Model	**Detection Accuracy (%)**	**False Positive Rate (%)**	**Execution Time (ms/sample)**
Isolation Forest	87.5	4.8	2.3
One-Class SVM	84.1	6.2	4.1
Local Outlier Factor	80.6	5.7	1.9
Autoencoder	89.4	3.9	5.5
River Model	78.2	7.4	1.1
Ensemble (Proposed)	93.8	2.1	3.7

8.1.3 Explainability via LLMs

Anomalies identified by the system are contextualized using a locally deployed Large Language Model (LLaMA 3 via Ollama). The model generates human-readable explanations for each flagged event, enhancing interpretability and user trust.

Table 8.3: Evaluation of LLM-Generated Explanations

Evaluation Metric	**Average Score (Out of 5)**
Explanation Clarity	4.7
Relevance to Operational Context	4.6
Usefulness for Non-Technical Users	4.8
Consistency Across Cases	4.5
Inference Time (per sample)	$\sim$1.2 seconds

These explanations have proven highly effective for both technical audits and user-facing insights, especially in environments lacking detailed logs or domain expertise.

8.1.4 Real-Time Visualization and Alert System

The system includes an interactive dashboard built using React and ReCharts. It provides real-time monitoring, categorization analytics, anomaly visualization, and explainability integration. Key features include:

- Time window filtering and event aggregation
- Operation-wise breakdown (Insert, Update, Delete, Rename)
- File category and severity-level filtering
- Export options in CSV and JSON formats
- Visualizations: time series graphs, heatmaps, bar charts

The dashboard has been benchmarked for responsiveness and usability, confirming its suitability for continuous monitoring environments.

8.2 Screen Shots

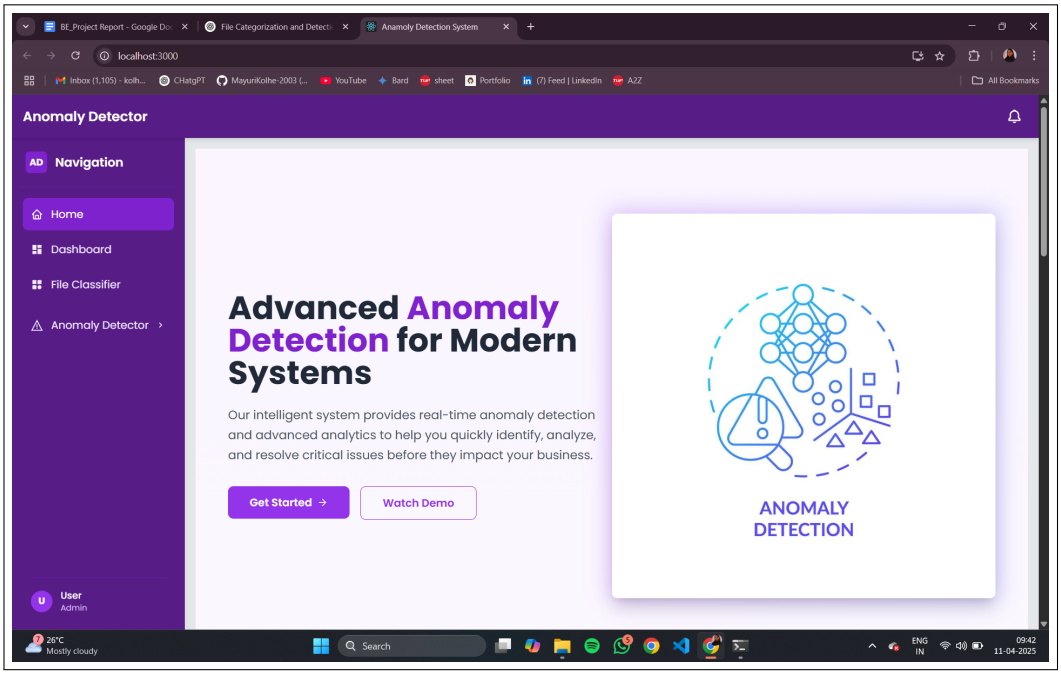


Figure 8.1: Home Page

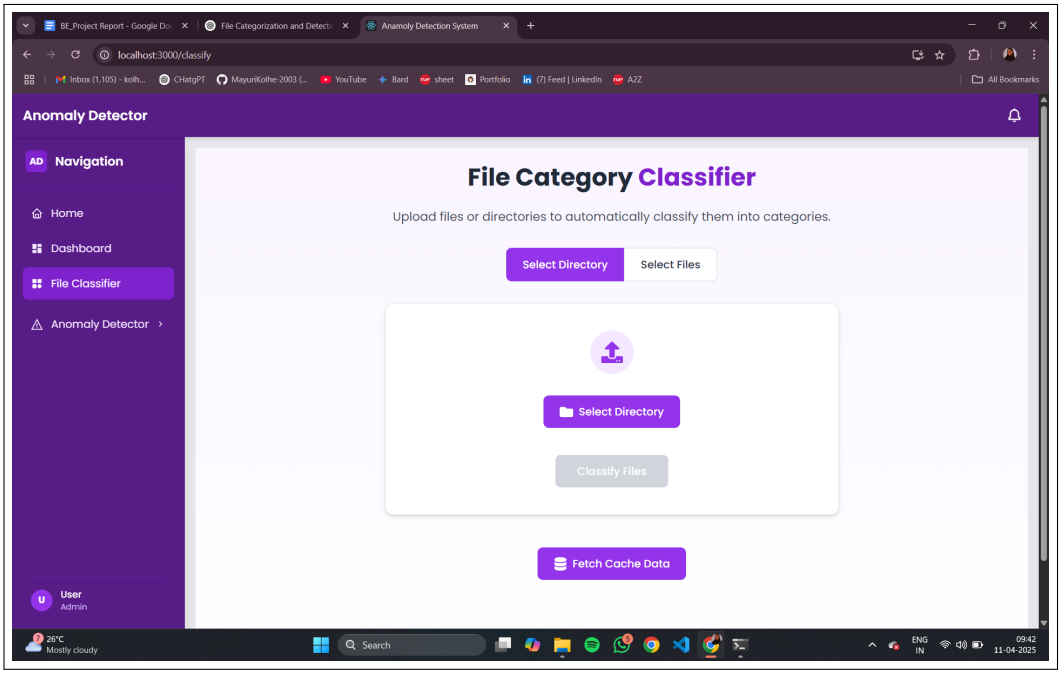


Figure 8.2: File categorizer

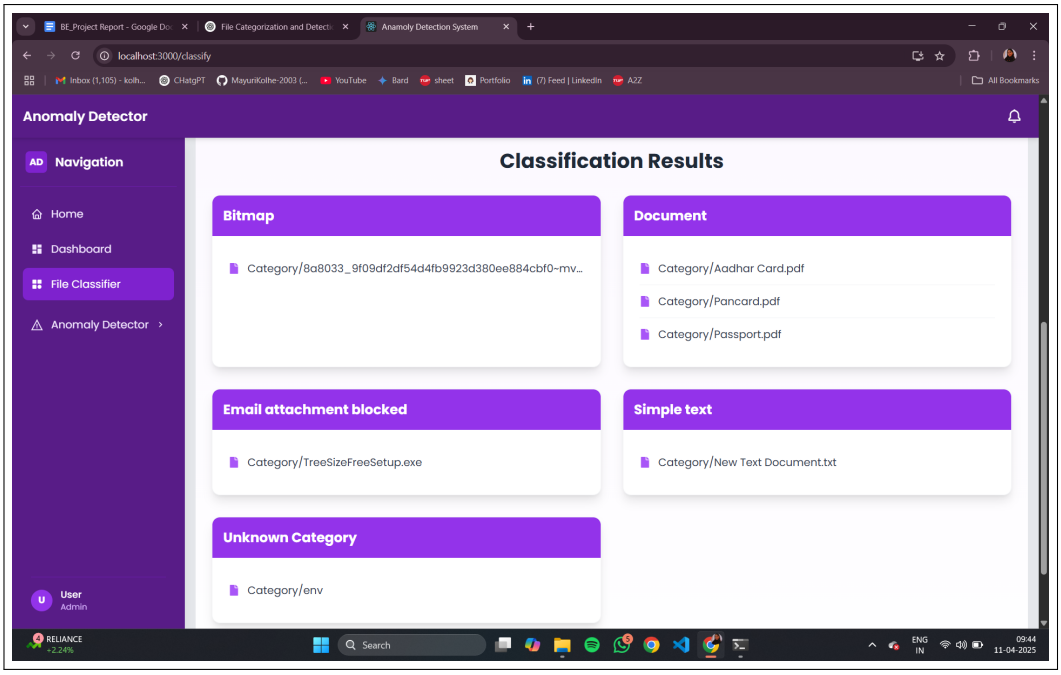


Figure 8.3: File categorizer results

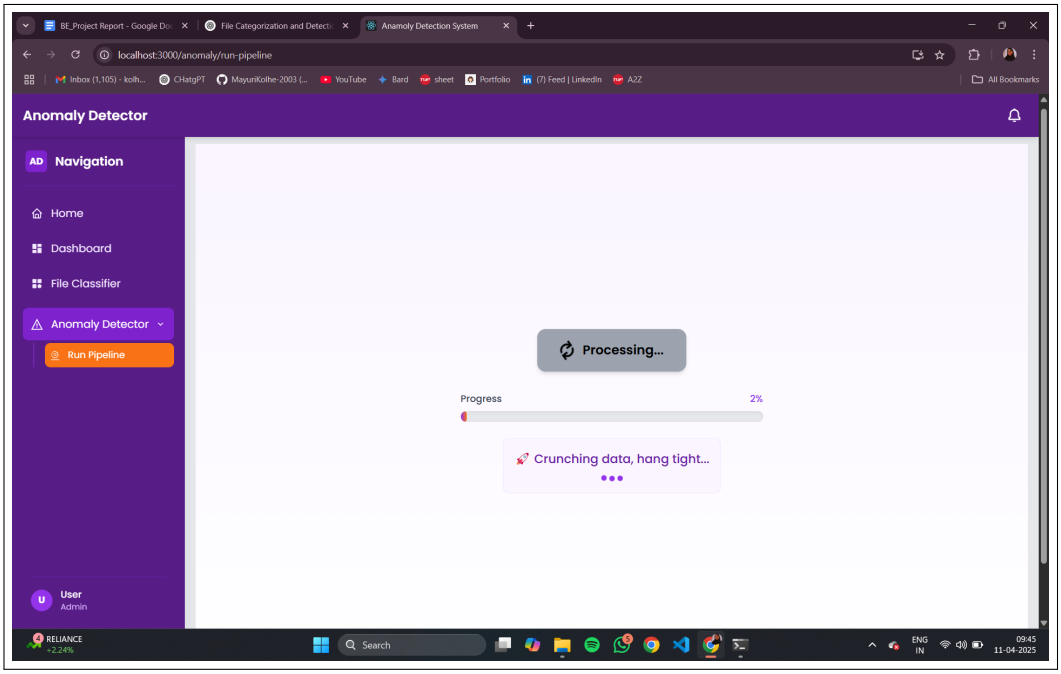


Figure 8.4: Pipeline to run Anomaly Detection

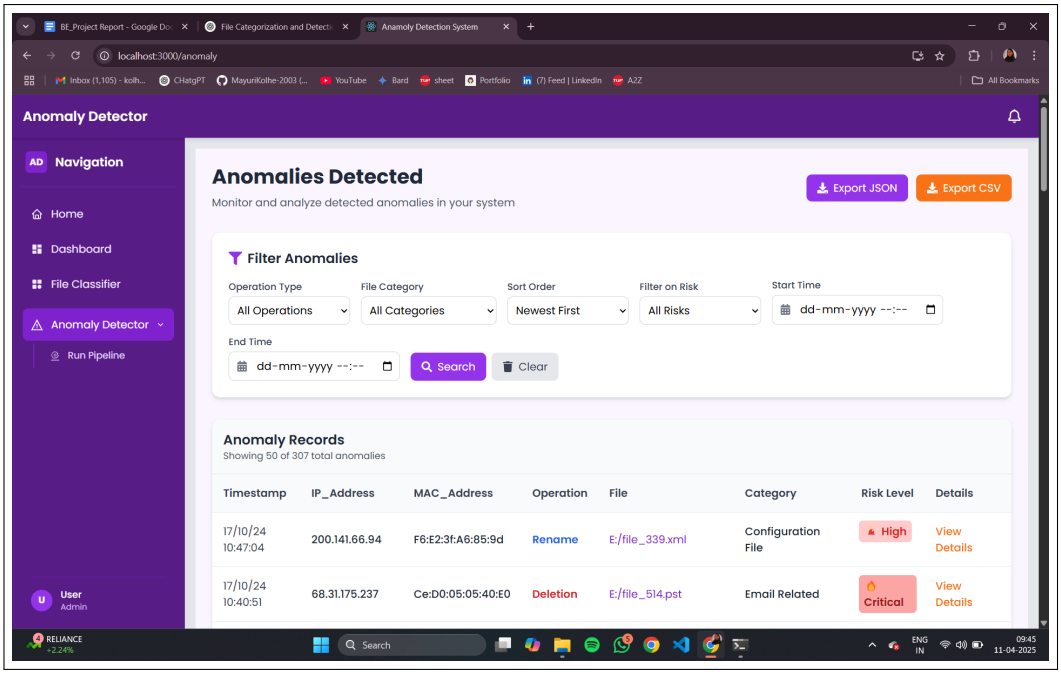


Figure 8.5: Detected Anomalies Dashboard

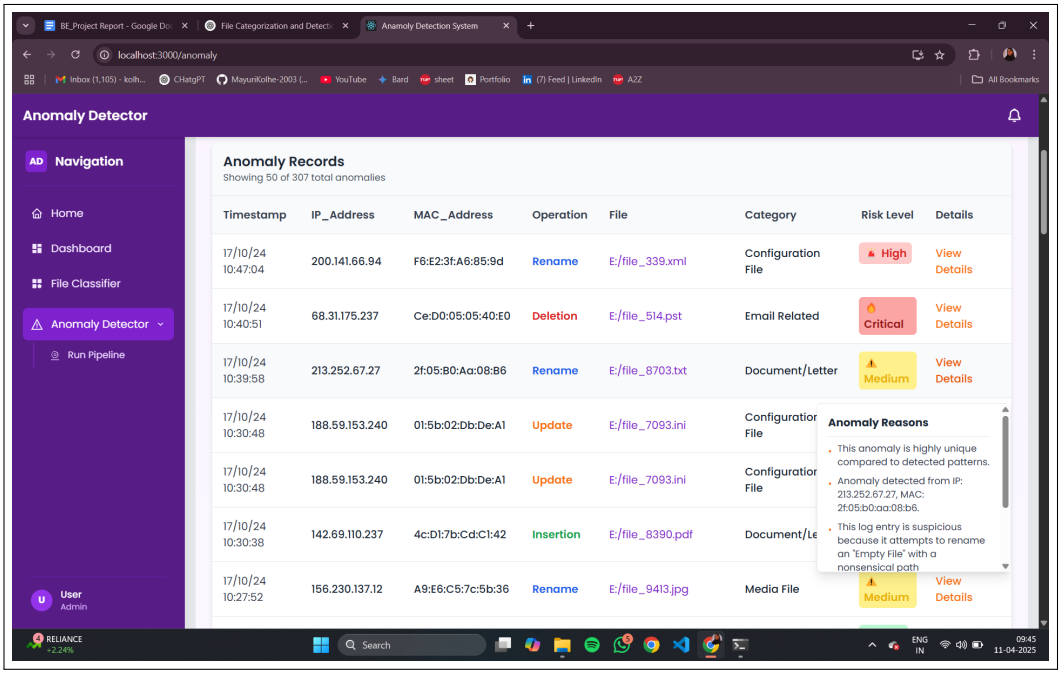


Figure 8.6: Detected Anomalies with reasons

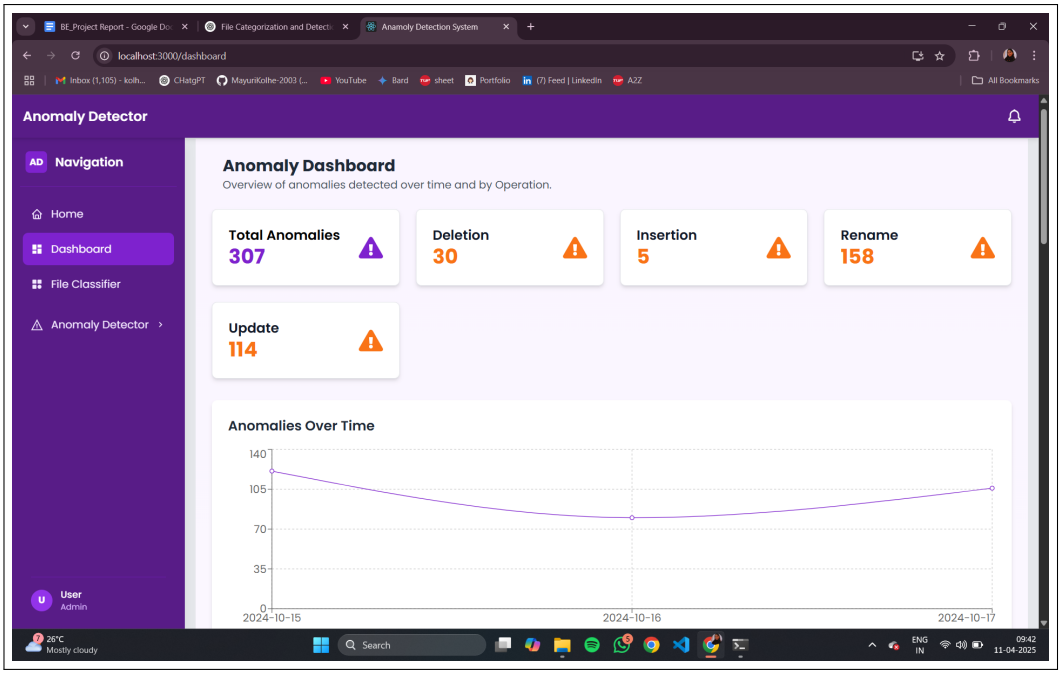


Figure 8.7: Dashboard for Analysis

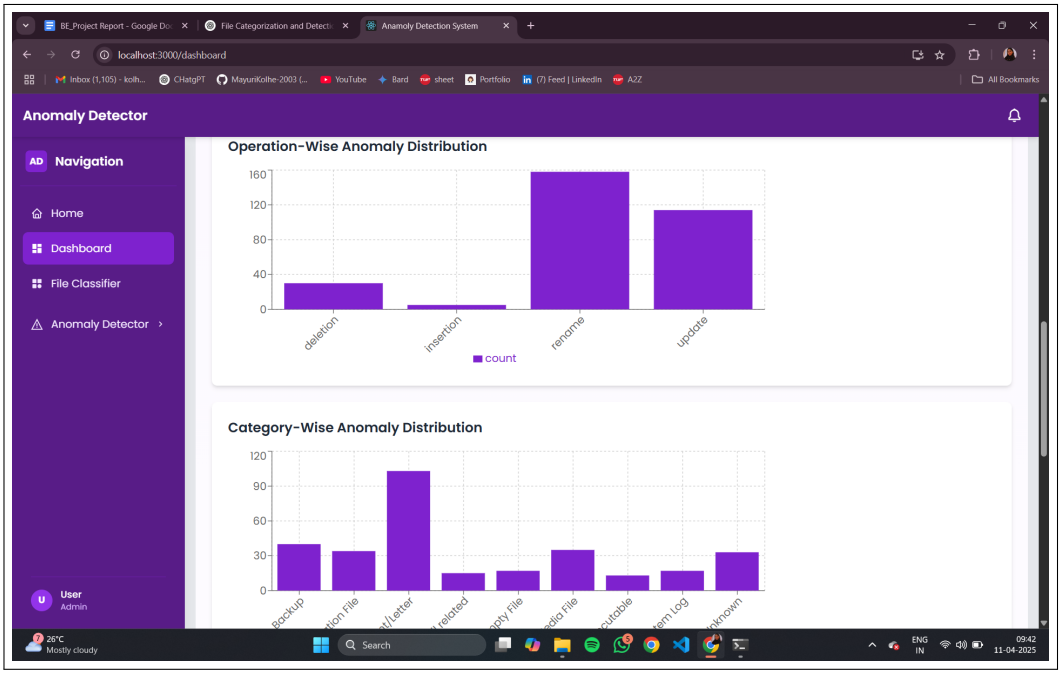


Figure 8.8: Anomaly distribution over operation and categories

CHAPTER 9

CONCLUSIONS

9.1 Conclusions

The project successfully delivers a comprehensive solution for automated file categorization and intelligent anomaly detection within file systems. By leveraging a combination of supervised and unsupervised machine learning techniques, the system accurately classifies files based on their extensions and content, while simultaneously monitoring behavior patterns to detect suspicious or abnormal activities in real-time. The integration of explainable AI models enhances user trust by providing clear, human-readable justifications for both classification and anomaly alerts. The real-time dashboard further contributes to the system's usability, offering intuitive visualization and actionable insights. Overall, the system not only improves file organization and operational efficiency but also significantly strengthens data security and governance, making it well-suited for deployment in enterprise-level and security-critical environments.

9.2 Future Work

The current system presents a robust foundation for intelligent file categorization, anomaly detection, and explainable monitoring. However, several enhancements can be pursued to improve adaptability, scalability, and system intelligence. Future development directions include:

9.2.1 User Behavior Profiling

To further improve anomaly detection precision, future work will involve modeling user- or role-specific behavioral baselines. By analyzing historical file operation patterns for individual users or departments, personalized anomaly thresholds can be defined, significantly improving the system's ability to detect insider threats or targeted malicious activity.

9.2.2 Real-Time Remediation Mechanisms

Future versions of the system may integrate automated response mechanisms that trigger remediation actions in real-time. For example:

- Blocking unauthorized file operations
- Quarantining suspicious files
- Triggering role-based access control checks
- Sending prioritized alerts to system administrators

This will transition the system from a passive monitoring tool to an active defense mechanism.

9.2.3 Continuous and Online Learning Models

Current models require periodic retraining to accommodate evolving file behaviors or new file types. Future iterations will integrate online learning and adaptive algorithms that incrementally update themselves based on new data streams, minimizing system downtime and reducing retraining overhead.

9.2.4 Multi-Platform and Cloud Integration

To expand the system's applicability, support for monitoring cloud-based file systems will be incorporated. This includes integration with platforms such as:

- Google Drive
- Dropbox
- Microsoft OneDrive
- AWS S3

Such extensions will allow unified anomaly detection across hybrid environments (local + cloud) and ensure seamless categorization and monitoring irrespective of storage location.

9.2.5 Enhanced LLM Personalization and Fine-Tuning

Future improvements to the LLM-powered explanation module include persona-aware fine-tuning and domain adaptation. This will enable explanations tailored to specific user roles (e.g., system admins vs. auditors) and sectors (e.g., education, healthcare, corporate), thereby improving interpretability and relevance.

9.3 Applications

9.3.1 Enterprise File Management Systems

Automates file organization and monitors unusual activities in corporate environments, helping IT teams manage data efficiently and detect internal threats.

9.3.2 Cybersecurity and Threat Detection

Identifies potential security breaches through behavioral anomalies, such as mass deletions or unauthorized access, aiding in early threat detection and prevention.

9.3.3 Cloud Storage Monitoring

Integrates with cloud platforms to classify uploaded files and monitor user activity, ensuring compliance and security in cloud-based infrastructures.

9.3.4 Digital Forensics and Audit Trails

Provides detailed logs and behavioral analysis that can be used in forensic investigations to trace suspicious operations or policy violations.

9.3.5 Data Loss Prevention (DLP) Systems

Helps organizations detect and respond to risky file operations, preventing accidental or malicious data leaks.

CHAPTER 10

APPENDIX

10.1 Appendix A

Problem statement feasibility assessment using, satisfiability analysis and NP Hard, NP-Complete or P type using modern algebra and relevant mathematical models.

This appendix evaluates the feasibility of the project titled *"A Dual-Phase Framework for Intelligent File Categorization and Real-Time Anomaly Detection in File Operation Behaviors"* by applying logical satisfiability, computational complexity theory, and modern algebraic modeling techniques.

1. Satisfiability Analysis

Phase 1 – File Categorization:
The problem is modeled as a deterministic classification task using file extensions and MIME types. For standard file types, the task is satisfiable and efficiently solvable using hash-based lookups and supervised models. Challenges arise with ambiguous or previously unseen extensions, which may require dynamic learning strategies.

Phase 2 – Anomaly Detection:
The task of identifying behavioral anomalies in file operations (e.g., mass deletions, renames) involves handling real-time data with inherent variability. While detection is satisfiable in small-scale scenarios, the problem becomes increasingly complex and less predictable at scale due to asynchronous events and overlapping operations.

2. Computational Complexity Classification

array

3. Application of Modern Algebra

Group Theory:
Applied to model categorization operations as transformations on file-

Task	**Complexity Class**	**Justification**
Basic File Categorization (Extension-based)	P (Polynomial)	Operates on hash mappings and decision trees; completes in deterministic time.
Handling Ambiguous/Mislabeled Files	NP-Complete	Requires combinatorial resolution and probabilistic inference.
Real-Time Anomaly Detection	NP-Hard	Involves dynamic data streams, multivariate analysis, and real-time constraints.

Table 10.1: Complexity Classification of Project Tasks

type sets, enabling structured classification logic with reversibility and consistency.

Graph Theory:
Used in anomaly detection to represent file behavior as a directed graph where:

- Nodes denote file states (e.g., created, modified, renamed).
- Edges represent operations, allowing traversal and anomaly scoring based on unusual patterns or cycles.

4. Conclusion

The proposed dual-phase system is partially feasible within defined boundaries:

- **File Categorization** is computationally tractable and robust using traditional algorithms and hash structures.
- **Anomaly Detection** introduces scalability and performance challenges due to its NP-Hard nature in real-time, dynamic environments.

The use of algebraic modeling adds formal structure, improving the interpretability and manageability of system behaviors. This analysis supports the viability of the framework, provided that optimizations,

heuristics, and scalable architectures are employed for the anomaly detection phase.

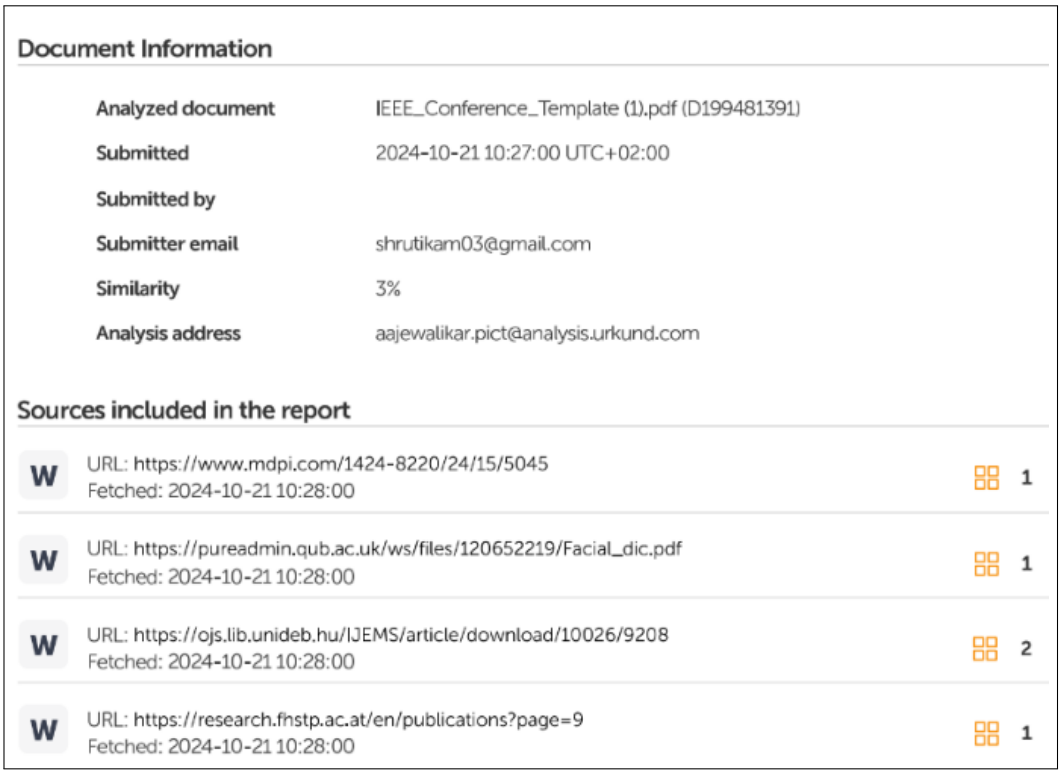
10.2 Appendix B

Details of paper publication: name of the conference/journal, comments of reviewers, certificate, paper.

- **Paper Title**: *Enhancing File Management Systems with AI: A Dual-Phase Approach to Categorization and Anomaly Detection*
- **Conference Name**: 9th International Conference on Control Communication, Computing and Automation (ICCUBEA 2025)
- **Track Name**: Information, Cyber and Network Security
- **Paper ID**: 338
- **Date of Submission**: March 31, 2025

10.3 Appendix C

Plagiarism Report of project report.



Document Information

Analyzed document	IEEE_Conference_Template (1).pdf (D199481391)
Submitted	2024-10-21 10:27:00 UTC+02:00
Submitted by	
Submitter email	shrutikam03@gmail.com
Similarity	3%
Analysis address	aajewalikar.pict@analysis.urkund.com

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CHAPTER 11

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